HEAVY ION INJECTION CHAIN OF NICA COLLIDER

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NICA heavy ion injection chain (beam transfers)

Beam transfer from HILAC to Booster

- HILAC-Booster transport channel;
- Booster injection system.

Beam transfer from Booster to Nuclotron

- Booster fast extraction system;
- Booster-Nuclotron transport channel;
- Nuclotron high energy beam injection system.

Beam transfer from Nuclotron to Collider

- Nuclotron fast extraction system;
- Nuclotron-Collider transport channel;
- Collider injection system.

Beam transfer from HILAC to Booster

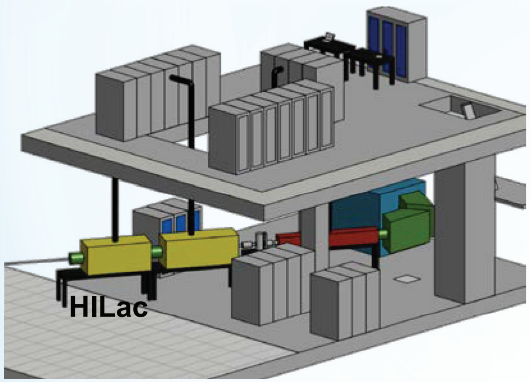
Beam parameters

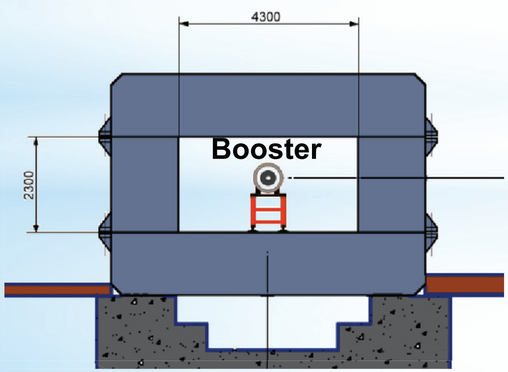
Ions	Au^{31+}
Energy, MeV/u	3.2
Magnetic rigidity, T m	1.6
Electric rigidity, MV	40
Ion number	$2 \cdot 10^9$

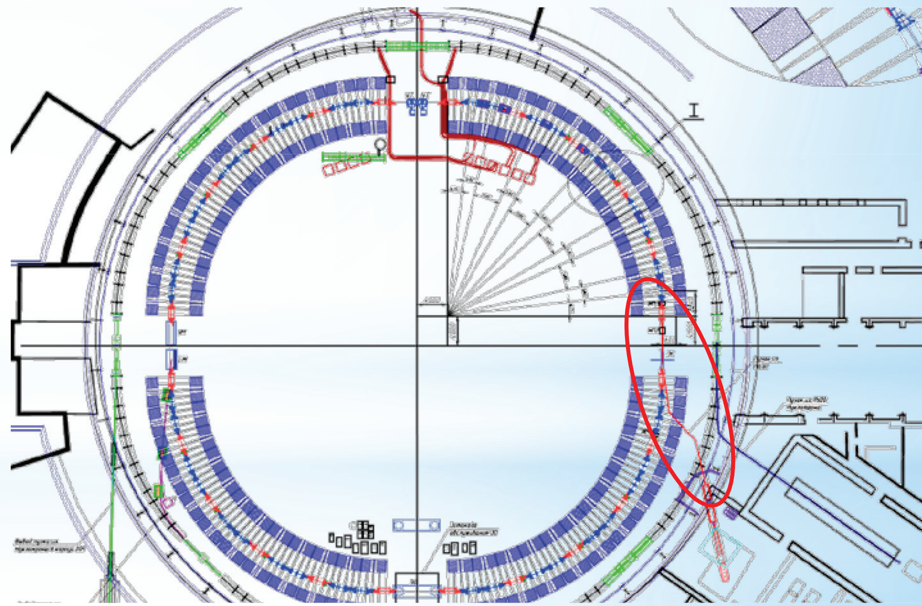
Goals

- Accumulation of required intensity of ions in Booster by means of several methods of beam injection.

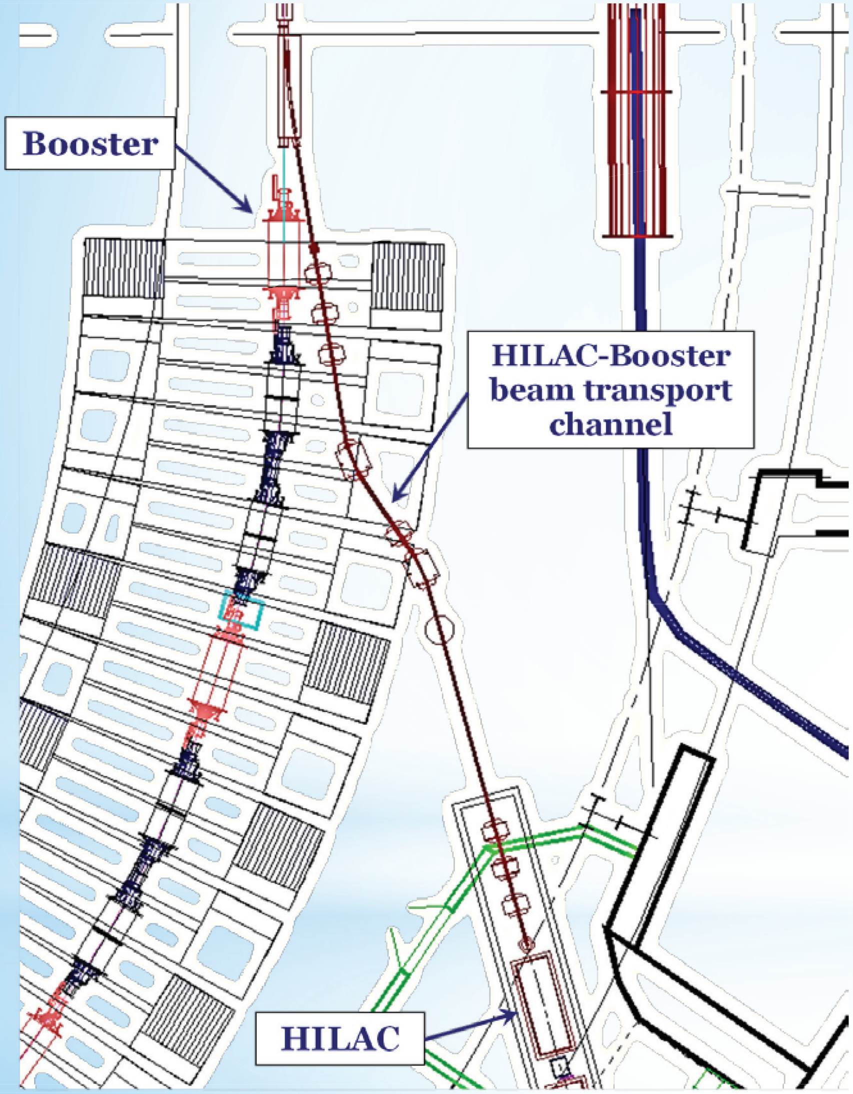
Commissioning: April 2019







HILAC-Booster beam transport channel



Main goals

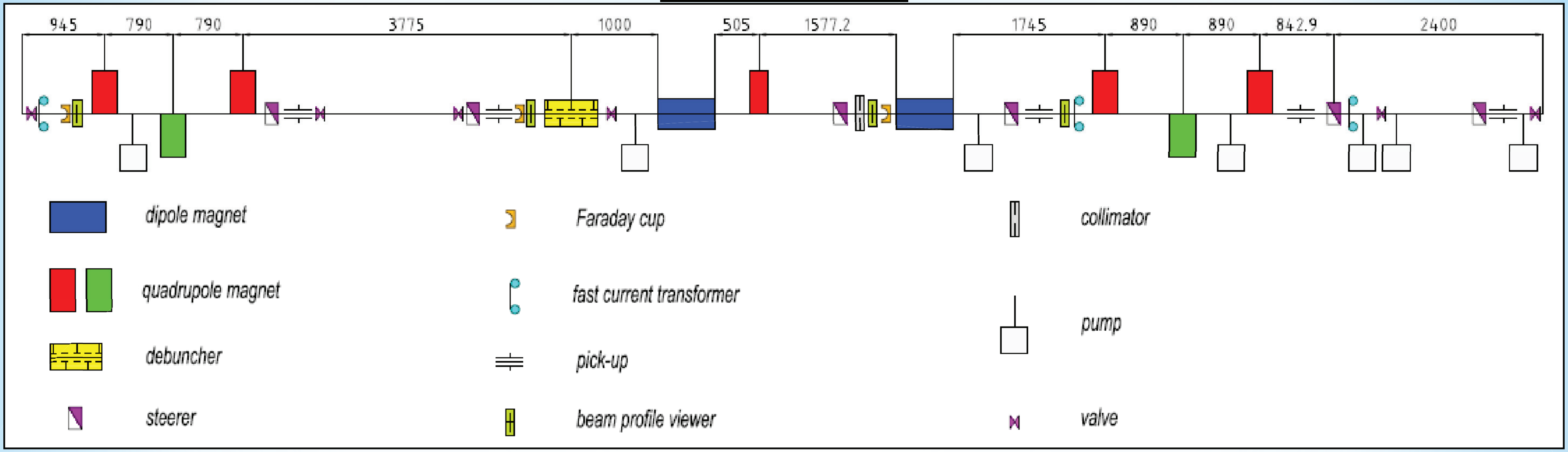
- The beam transport with minimal ion losses.
- The beam debunching.
- The beam matching.
- Separation and adsorption of neighbor charge states of ions.
- Providing different schemes of the beam injection into the Booster.

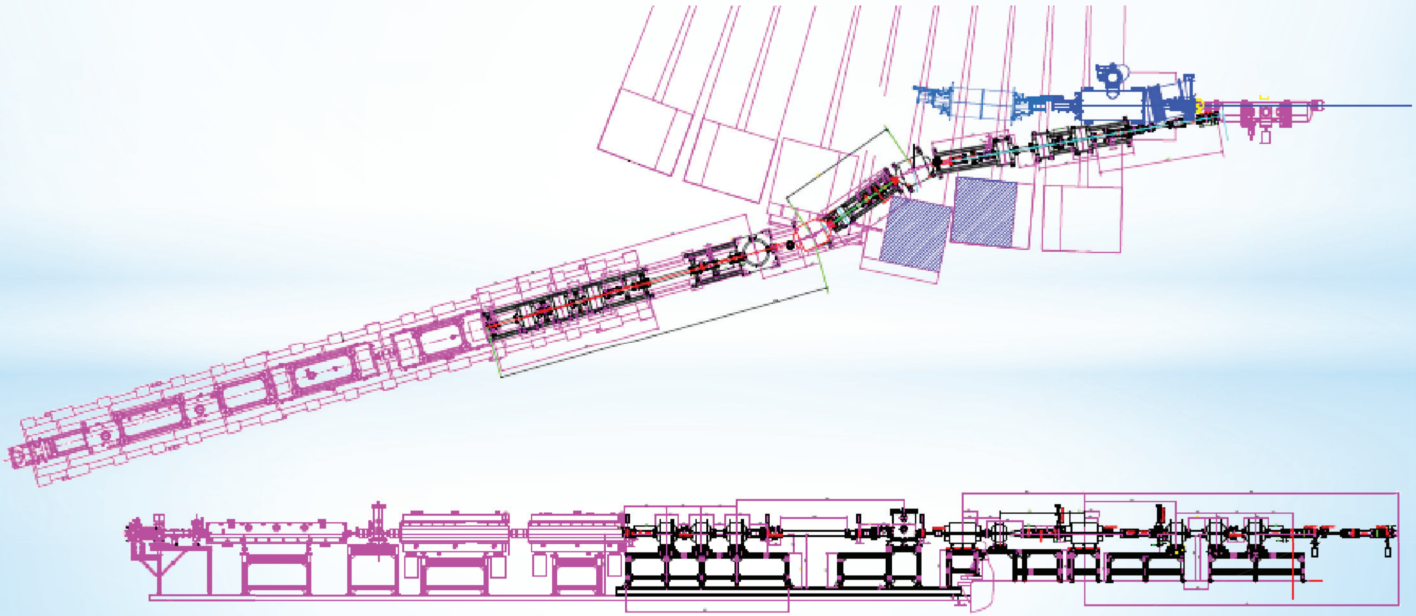
Features

- Beam transport in the median plane of the Booster.
- The debuncher in non-dispersive region of the beam transport channel.

HILAC-Booster beam transport channel

Composition

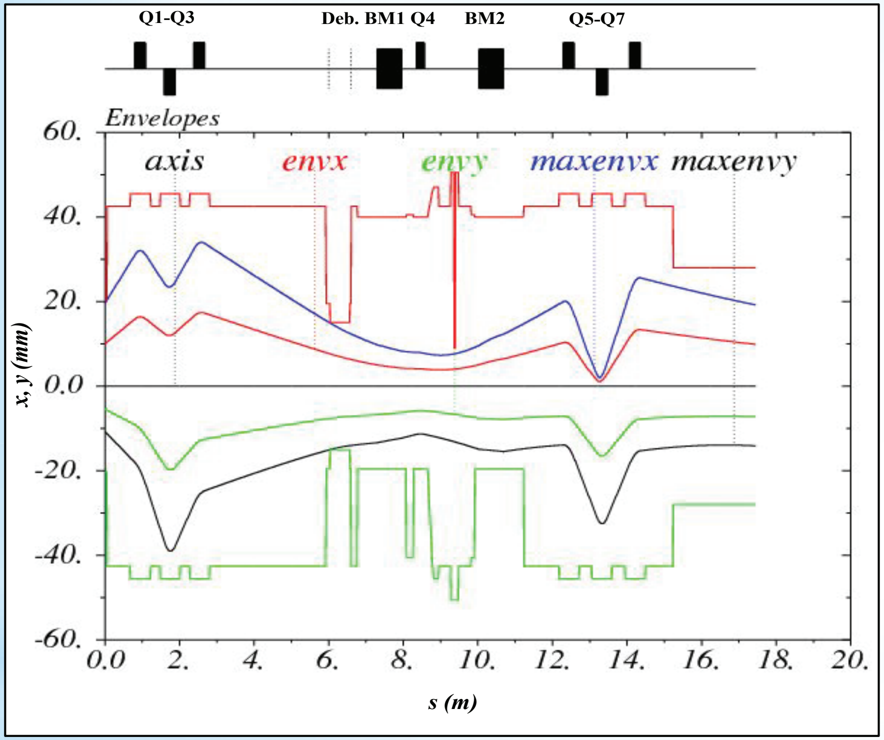




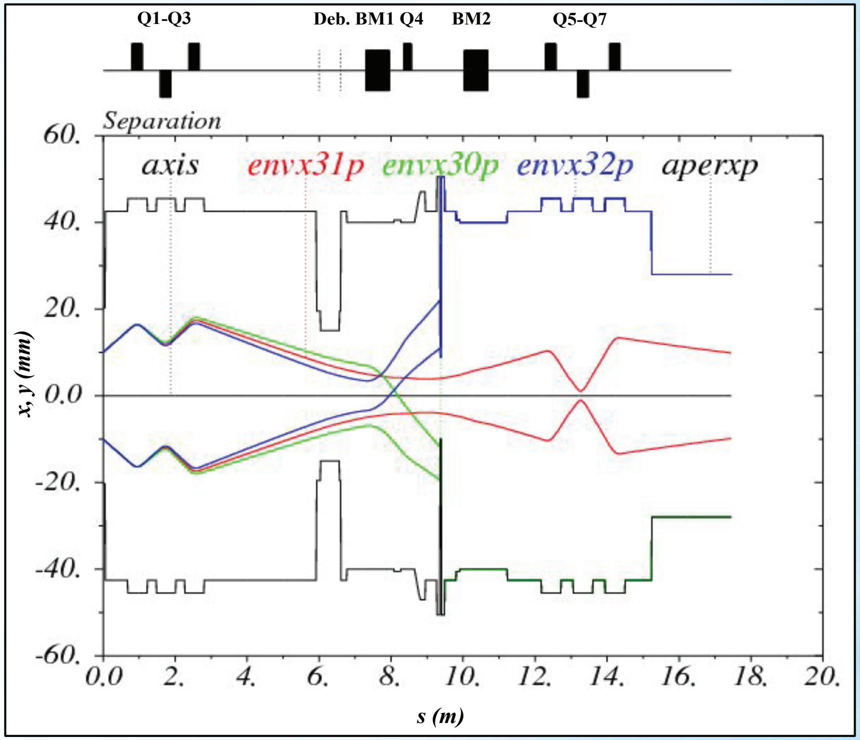
HILAC-Booster beam transport channel

Beam dynamics

Beam envelopes



Separation



- Horizontal and vertical acceptances of the channel are equal to 38 π·mm·mrad.
- Reducing $\frac{\Delta p}{p}$ from $5 \cdot 10^{-3}$ to $1 \cdot 10^{-3}$ by means of the beam debunching.

HILAC-Booster beam transport channel

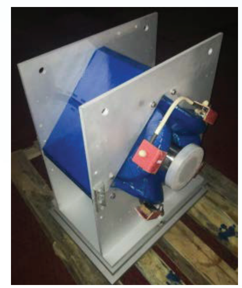
Parameters of main elements

Dipoles

Effective length, m	0.65
Max field, T	1
Gap, mm	45

Quadrupoles

Effective length, m	0.29 / 0.21
Max gradient, T/m	10 / 19
Gap (diameter), mm	95 / 71



Steerers

Effective length, m	0.25
Max field, T	0.05
Gap, mm	125 × 110

HILAC-Booster beam transport channel

Parameters of main elements

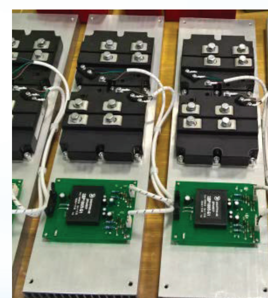
Debuncher

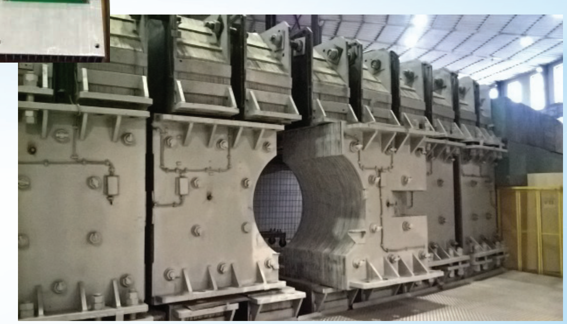
Inner length, m	0.49
Frequency, MHz	100.625
Max effective voltage, kV	260

Designed and fabricated by Bevatech team

Status of channel equipment

- Power supplies are being assembled.
- Beam diagnostics is purchased partially. Full set of beam diagnostics will not be purchased until the channel commissioning.
- First part of vacuum equipment is delivered. Full set of vacuum equipment will be delivered in the nearest future.
- Design of stands is near completion.





Beam transfer from Booster to Nuclotron

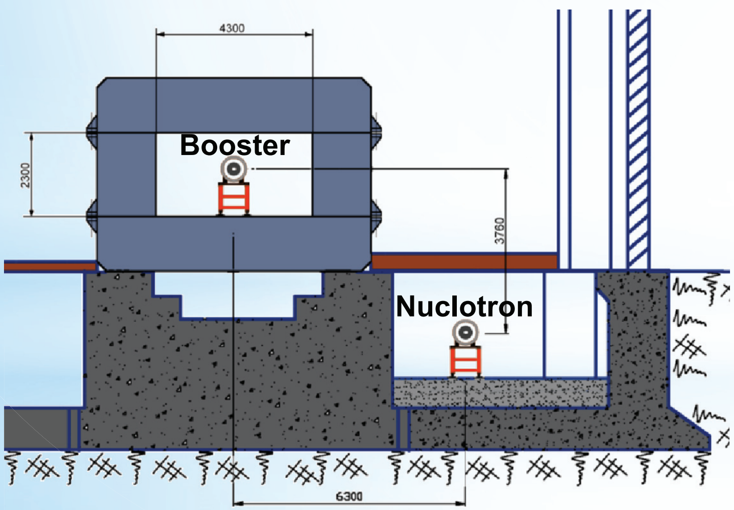
Beam Parameters

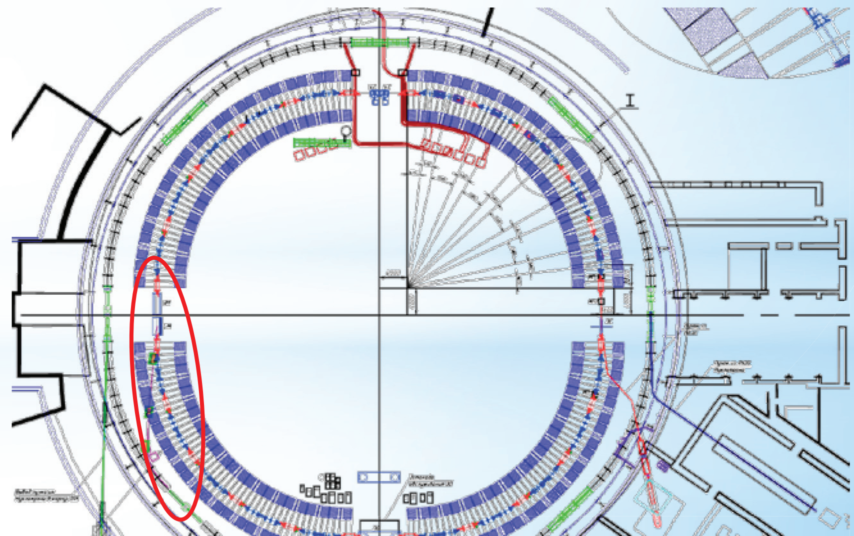
Ions: before stripping station after stripping station	 Au^{31+} Au^{79+}
Maximum magnetic rigidity of ions, T m: before stripping station after stripping station	 25 10
Ion number	$1.5 \cdot 10^9$

Goals

- Transfer of the beam with parameters which can be altered in wide ranges due to 1) use of different schemes of beam injection into Booster and 2) use of an electron cooling system.
- Ion stripping to a maximum charge state.

Commissioning: December 2019



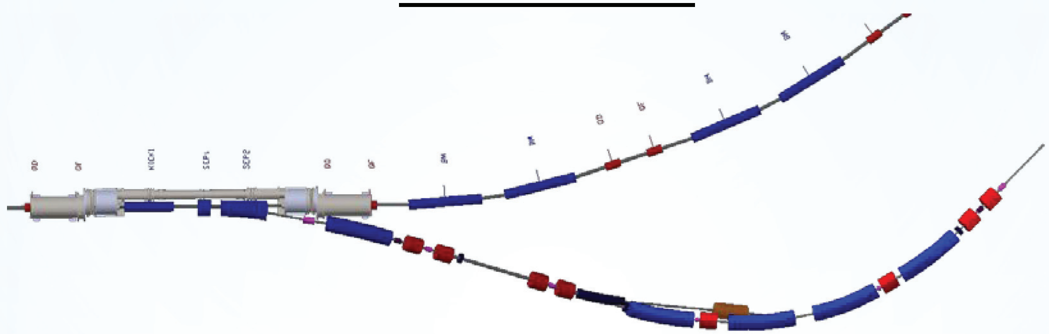


Booster-Nuclotron beam transport channel

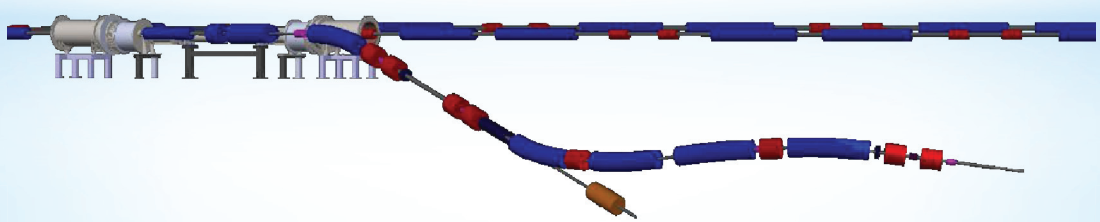
Goals

- The beam transport with minimal ion losses.
- Separation of neighbor charge states. Estimates of ion stripping at energy of 580 MeV/u: 100% Au^{31+} $\rightarrow$ 80% Au^{79+}, ~20% Au^{78+}. Due to high intensity of Au^{78+} ions they have to be extracted from the channel to an absorber.

View from above

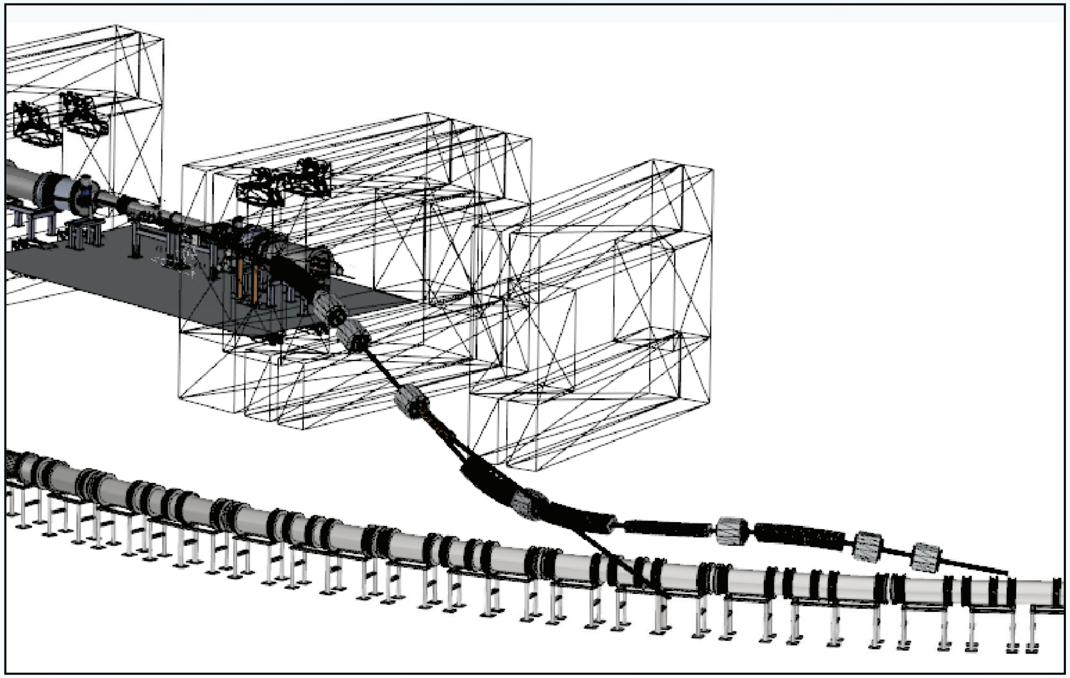


Side view

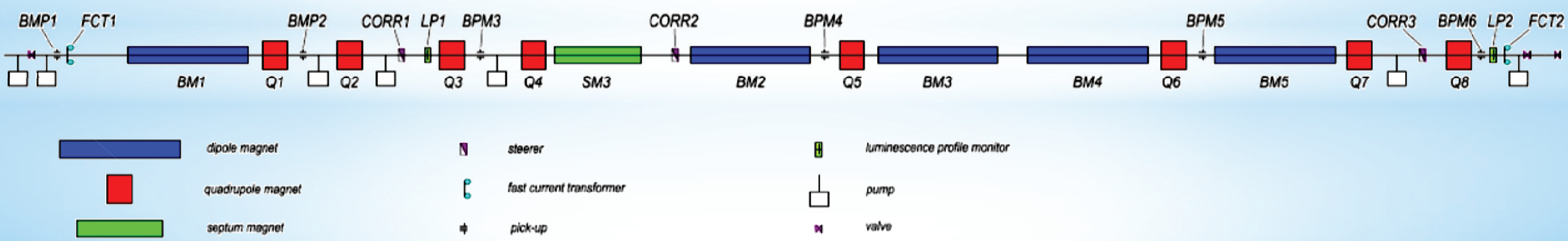


Booster-Nuclotron beam transport channel

3D model

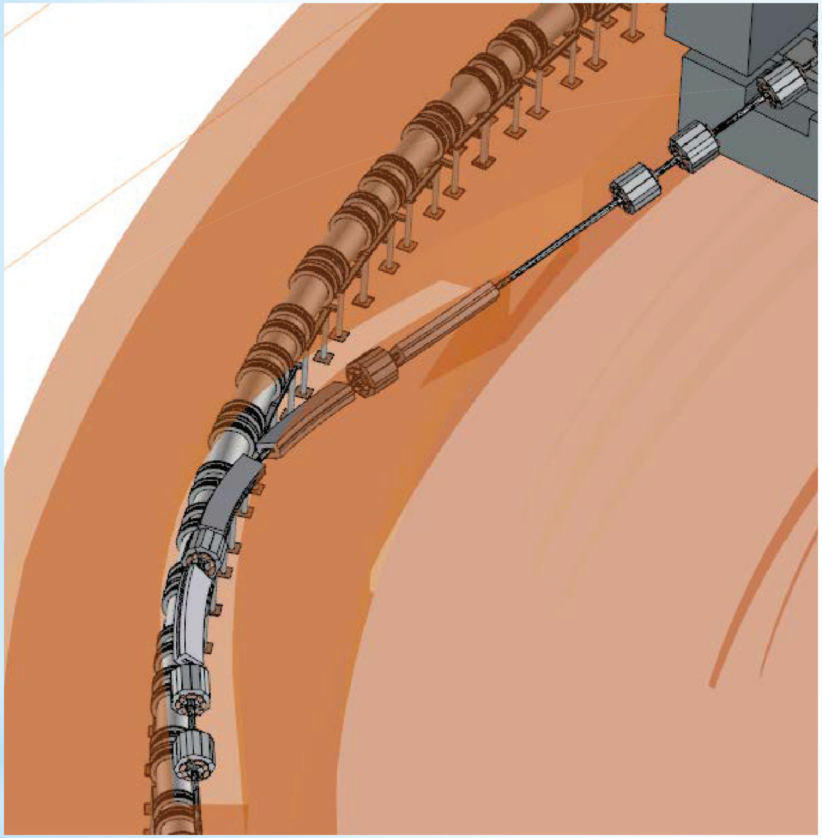


Composition



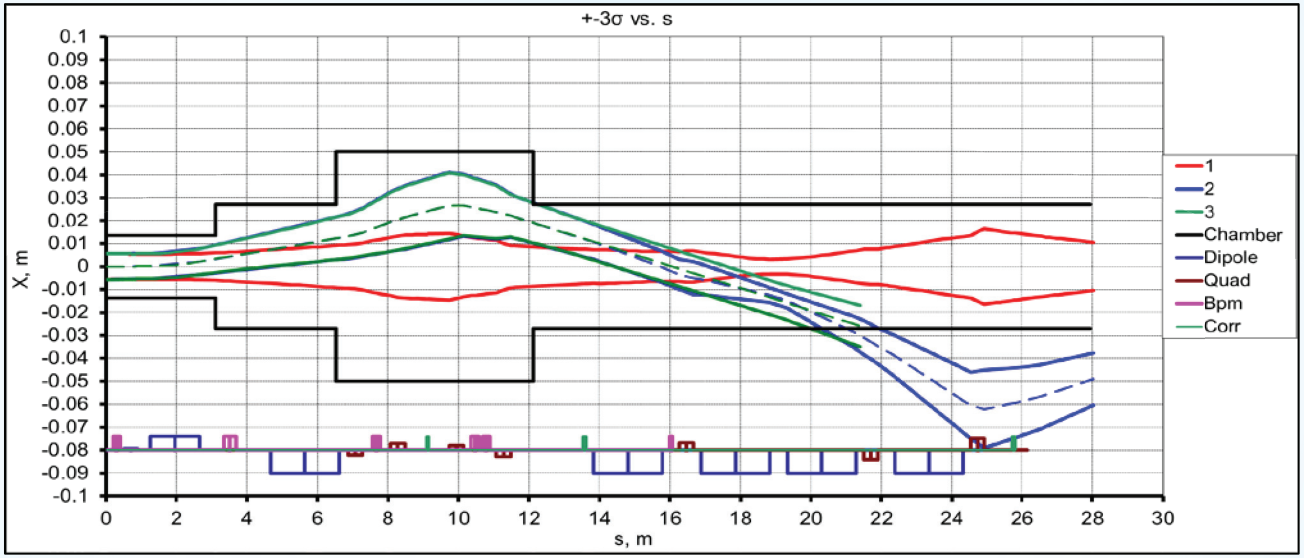
Booster-Nuclotron beam transport channel

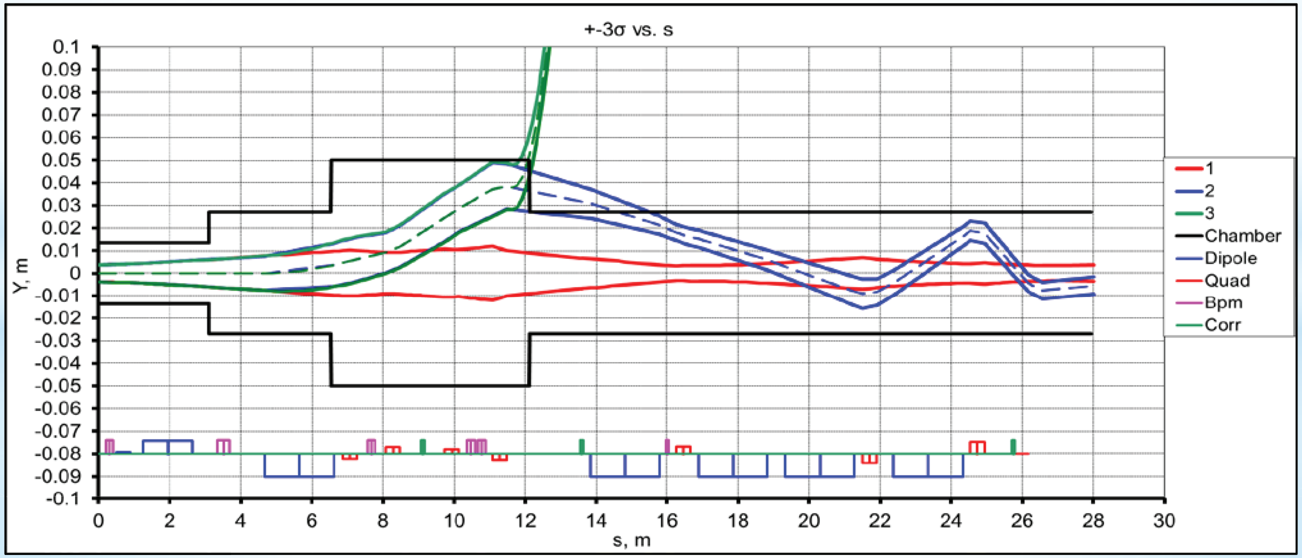
Layout



Booster-Nuclotron beam transport channel

Beam dynamics





Booster-Nuclotron beam transport channel

Parameters of magnetic elements

Magnetic element	Type	Effective length, m	Max. magnetic field (gradient), T (T/m)
BM1 - BM5	sector dipole	1.95	1.58
SEPT_2	septum	1.4	1.3
Q1 - Q4	quadrupole	0.4	14
Q5 - Q8	quadrupole	0.4	20
Corr1 - Corr3	steerer	0.1	0.15

Being designed by BINP team

Beam transfer from Nuclotron to Collider

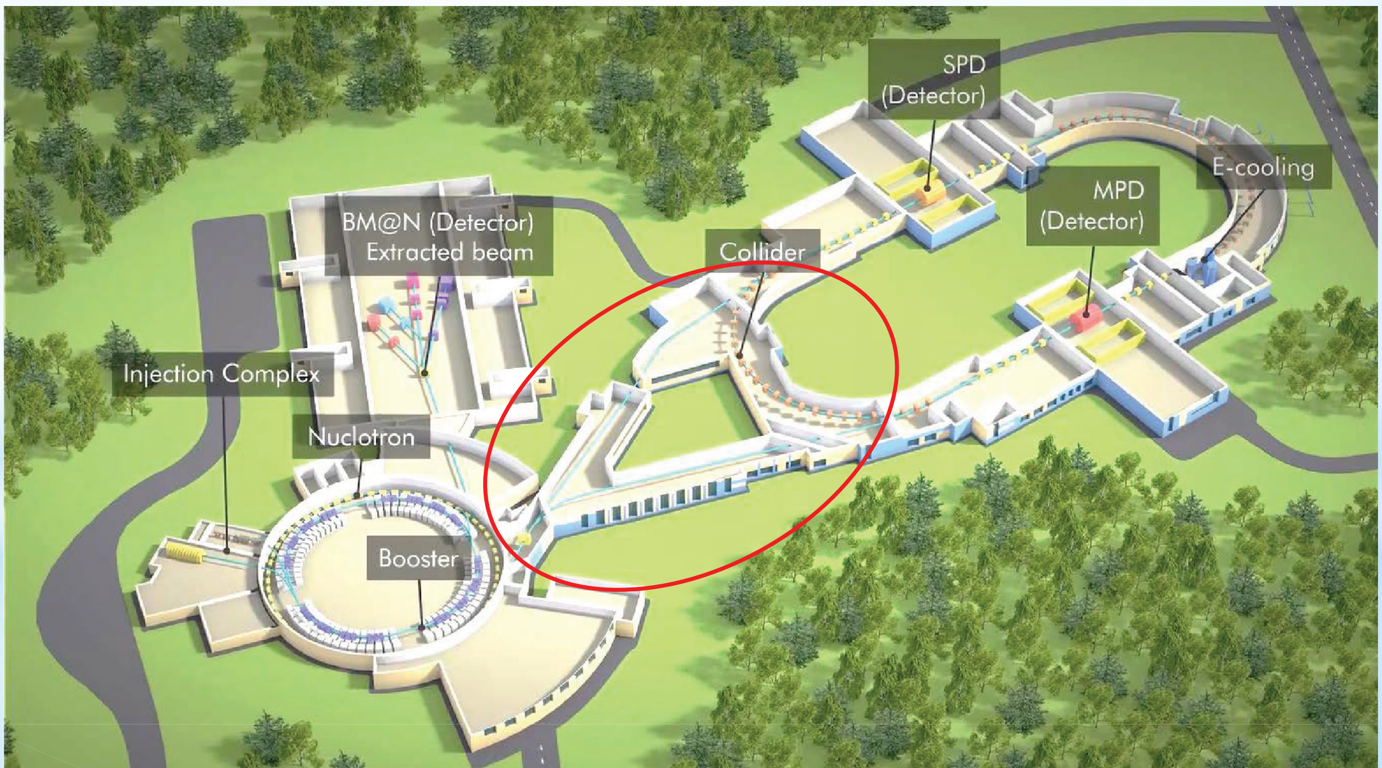
Goals

- Accumulation of required intensity of ions (with help of barrier bucket system and beam cooling systems of Collider).

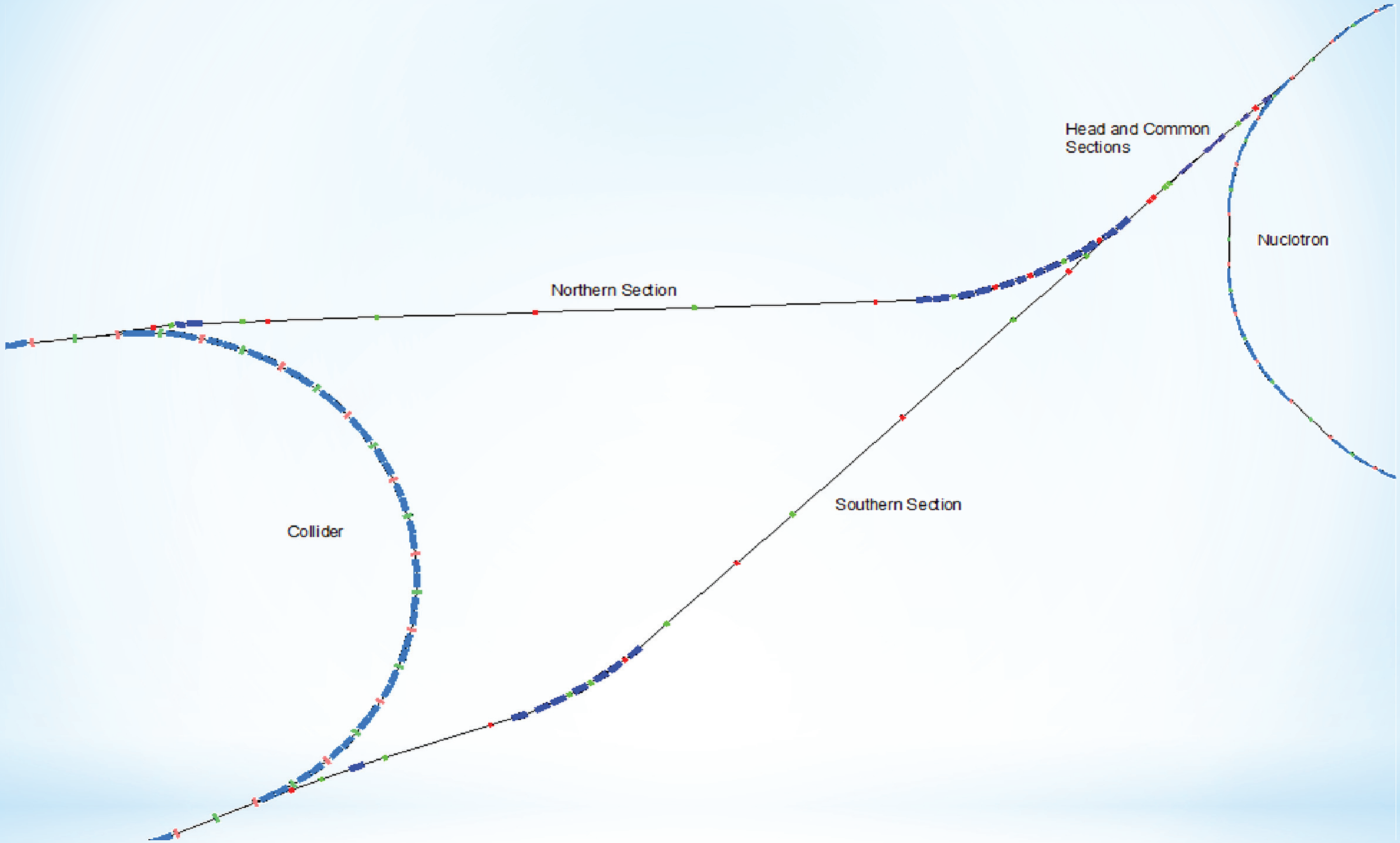
Commissioning: December 2020

Beam Parameters

Ions	Au^{79+}
Energy of ions, GeV/u	1 ÷ 4.5
Magnetic rigidity of ions, T m	14 ÷ 45
Ion number	$1 \cdot 10^9$



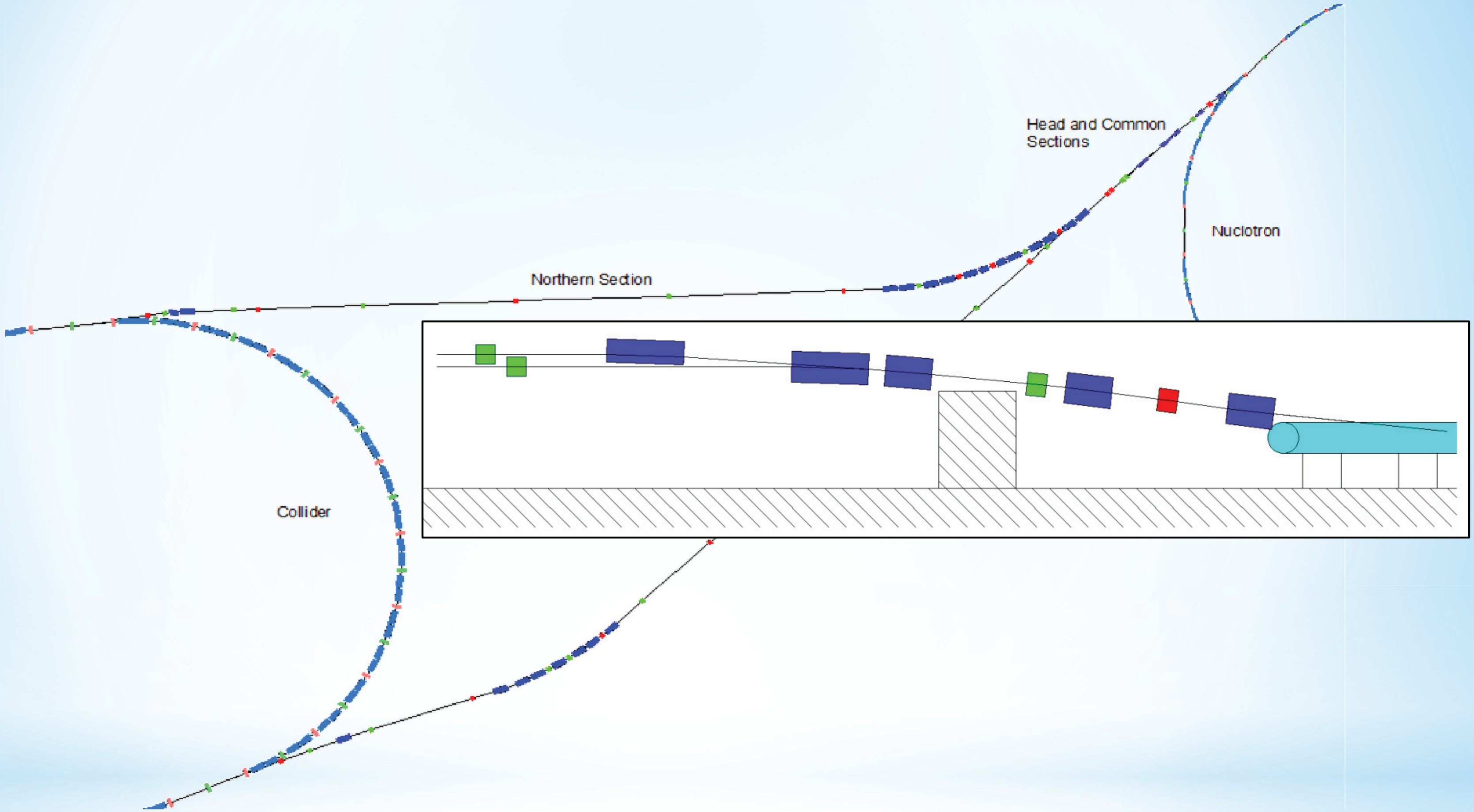
Nuclotron-Collider beam transport channel



Goals

- The beam transport with minimal ion losses.
- The beam matching with lattice functions of the Collider rings (except vertical dispersion).

Nuclotron-Collider beam transport channel



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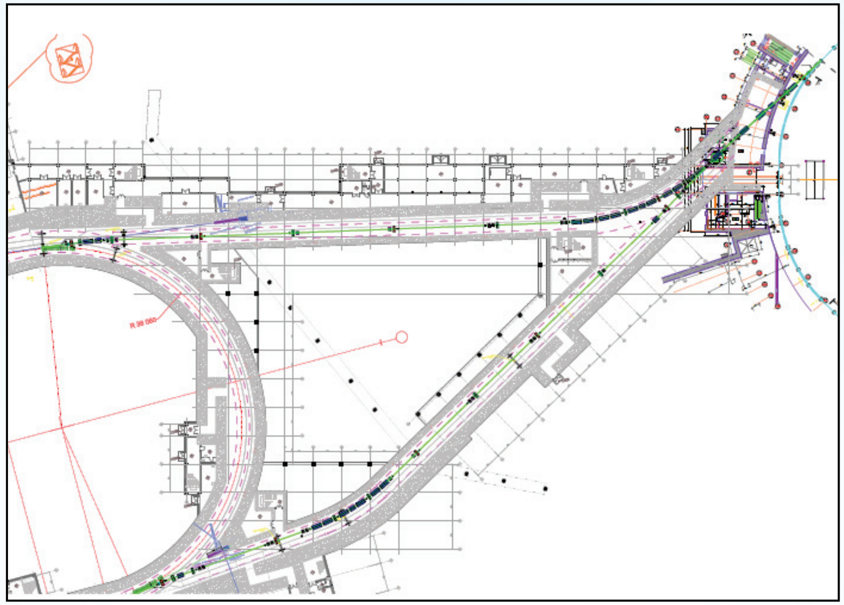
Nuclotron-Collider beam transport channel

Layout



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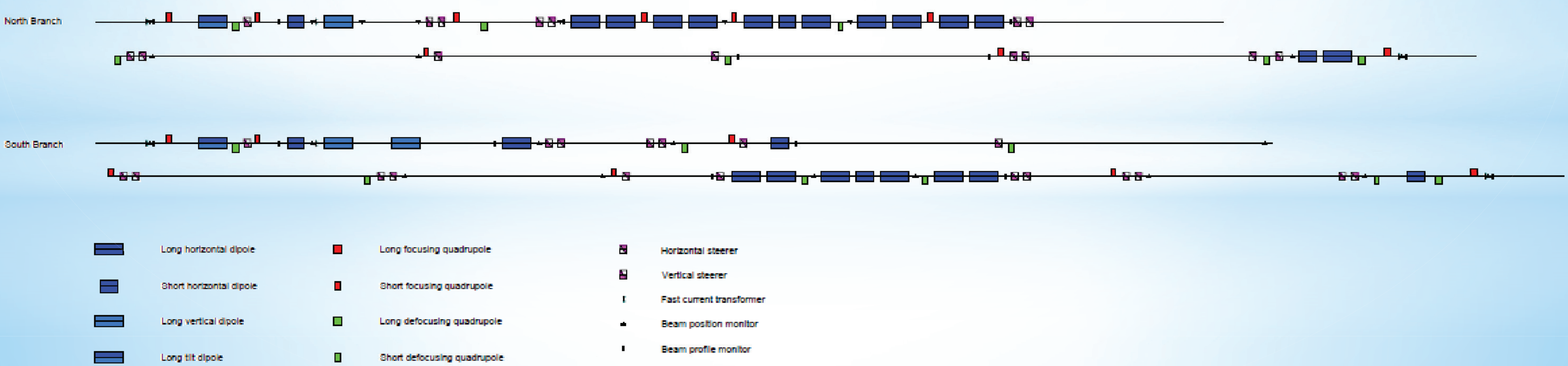
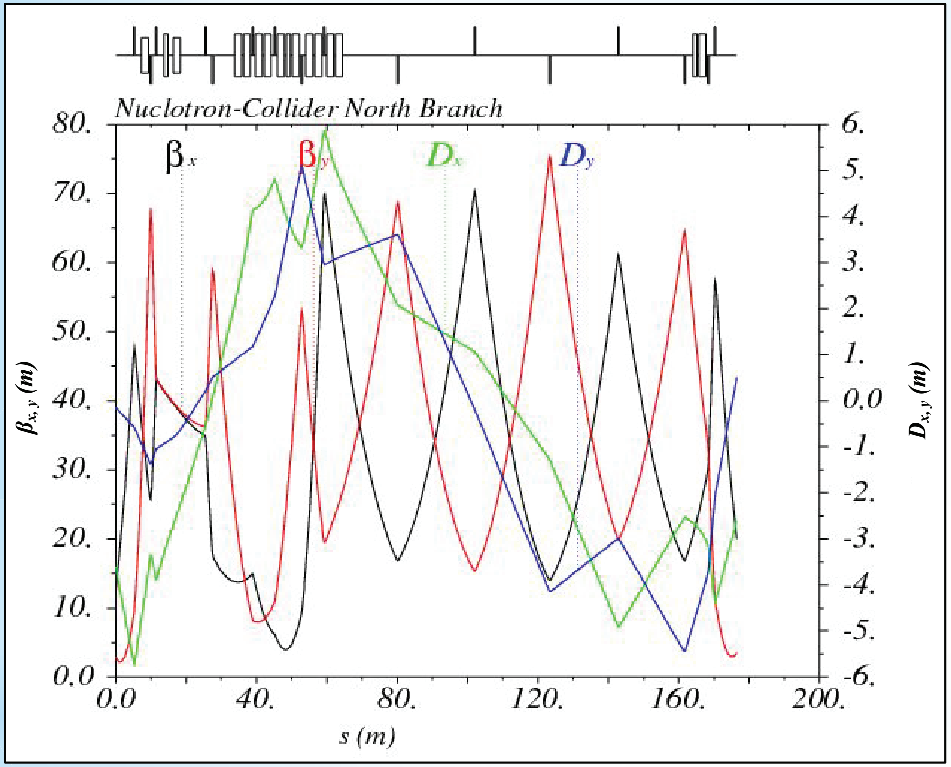
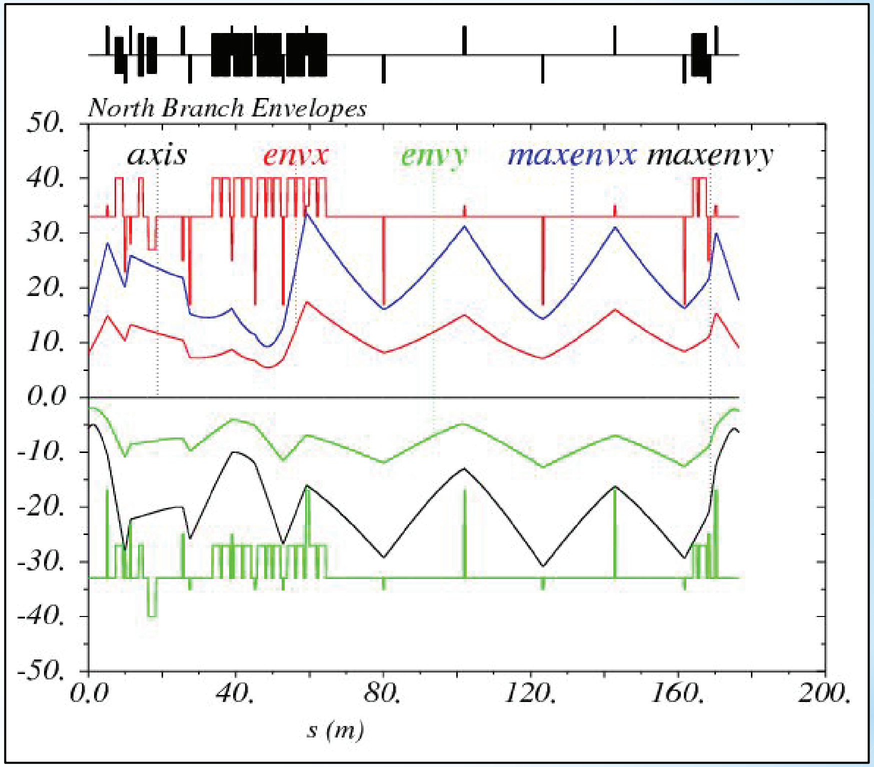


Nuclotron-Collider beam transport channel

Beam dynamics (North Branch)

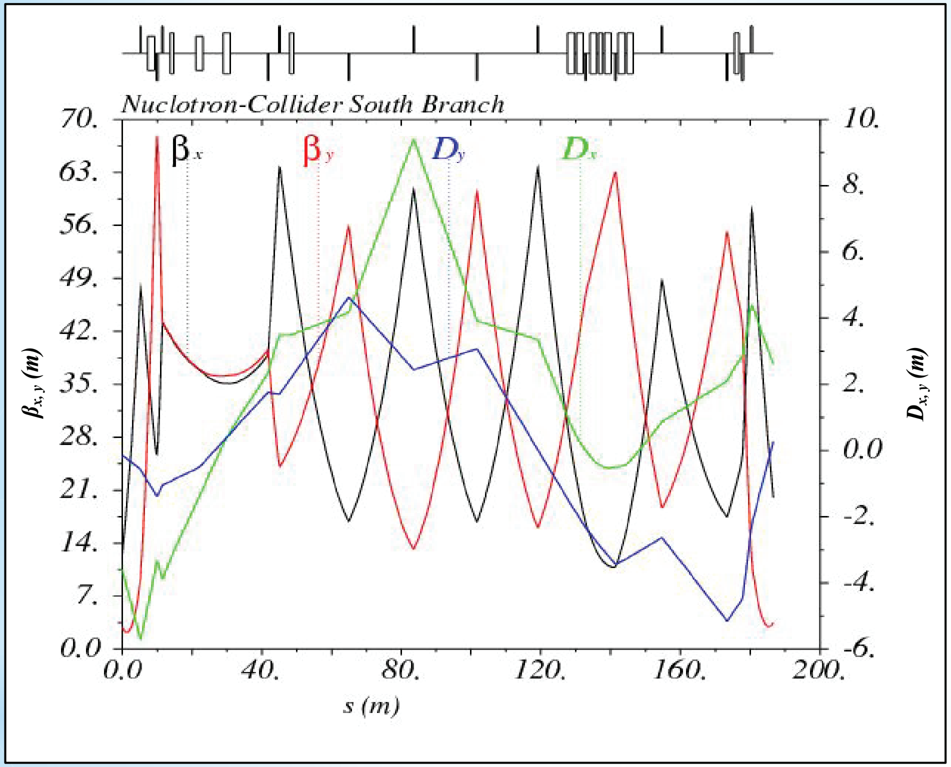


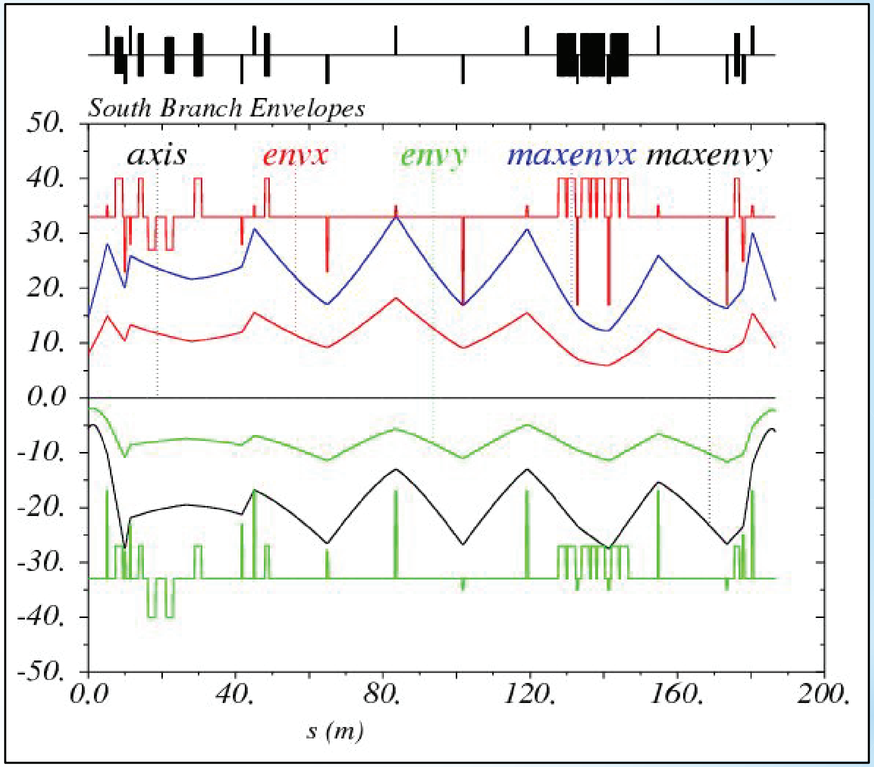


Horizontal and vertical acceptances of the north branch of the channel are equal to 13 and 10 π·mm·mrad correspondingly.

Nuclotron-Collider beam transport channel

Beam dynamics (South Branch)



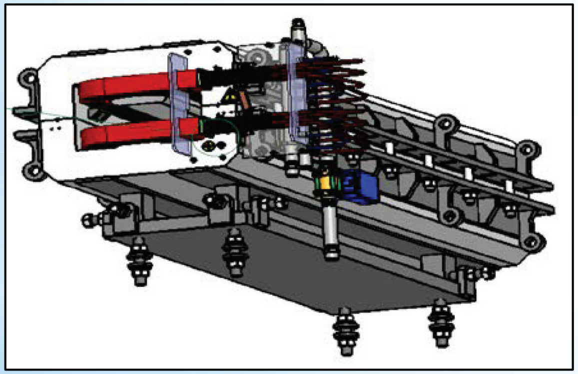


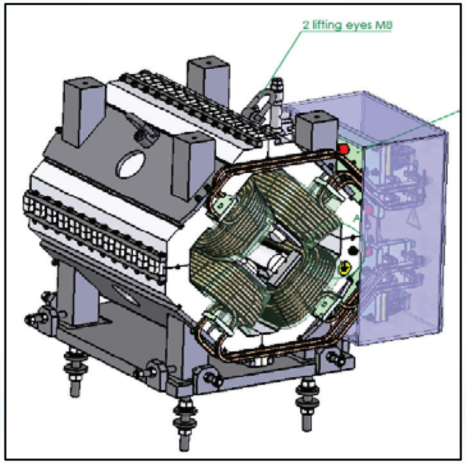
Horizontal and vertical acceptances of the south branch of the channel are equal to 13 and 10 π·mm·mrad correspondingly.

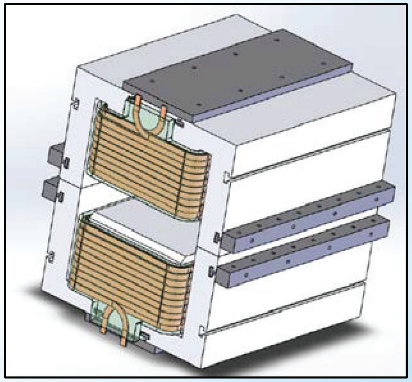
Nuclotron-Collider beam transport channel

Parameters of pulsed magnet elements

Magnetic element	Number	Effective length, m	Max. magnetic field (gradient), T (T/m)
Long dipole	21	2	1.5
Short dipole	6	1.2	1.5
Quadrupole Q10	22	0.353	31
Quadrupole Q15	6	0.519	31
Steerer	33	0.466	0.114







Designed by SigmaPhi

THANK YOU FOR ATTENTION